GRIDRUSH

speed-to-power agent for grid-friendly compute · MW energized in months, not years

SF × MIT Hack Day: Energy · AGI House SF · 8 Nov 2026
tracks 1 + 2 + 3 (sites · grid capacity · load management)

A data center can be built in **2–3 years**, but connecting it takes **7–10 years** (PG&E, on stage). ERCOT tracks **~410 GW** of large-load requests, ~87% data centers, against **7.5 GW** approved to energize since 2022. GRIDRUSH answers the question the room asked: how much compute can actually come online if a load accepts a **priced reliability commitment** and curtails only in the few hours the grid is stressed?

NET VALUE VS BEST HONEST PLAN
4.3×
$279.9M vs $65.1M

MW ONLINE AT MONTH 12
85.4
firm-only plan: 0 MW

SELLABLE COMPUTE, 36 MO
2491
MW·months, SLA met · 32/40 sites

VIOLATIONS / SLA BREACH
0
independent hour-by-hour check

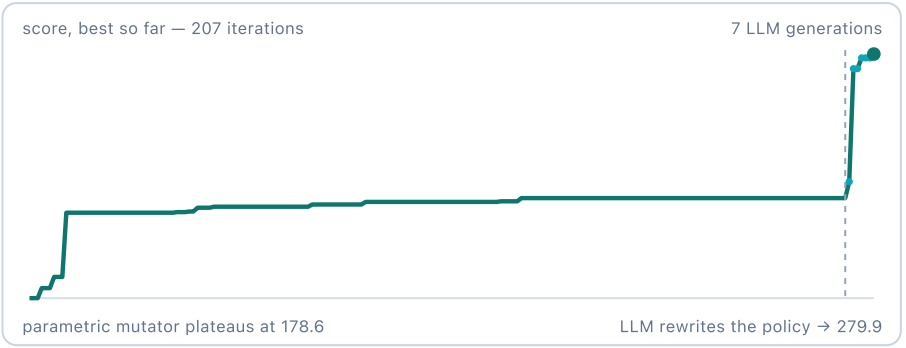
THREE STRATEGIES, SAME 40 SITES, REAL 2025 HOURLY REPLAY

STRATEGY	MW @ MONTH 12	SELLABLE MW·MONTHS	NET VALUE, 36 MO
All firm — wait for the queue	0.0	651	$65.1M
All non-firm at max MW, no discipline	172.6	1448 of 3384 gross	$-449.5M
GRIDRUSH agent	**85.4**	**2491**	**+$279.9M**

The agent does not pick one availability level globally: **16 sites at 99.99%** (revenue premium ×1.25), **6 at 99.9%**, **10 at 99%** where the grid is congested — buying 5–10 months of earlier revenue. It connects **32 of 40 sites** and refuses the rest. At the evening peak the naive plan must shed 28.4 MW; GRIDRUSH never sheds more than 1.0 MW.

THE DECISION IT MADE — batteries lose to the contract (they only buy availability the flexible SLA already provides), so the agent buys none

WHAT	FIRM	FLEXIBLE	FLEXIBLE + BATTERY	REFUSED
sites	0	32	0	8
SLA commitment	—	16 × 99.99% · 6 × 99.9% · 10 × 99%	—	—
months to energize	18–36	5–12	+1–3	—



HOW IT LEARNS

proposer → plan → 8,760-hour verifier → score → keep if better. Two proposers: a 9-parameter policy mutator, then a per-site local search where every proposal is judged by the full-year simulation. The best plan is then rewritten by an **LLM agent** that writes policy code: 7 generations, one recorded failure, one jump from 178.6 to 279.9. The policy also **repairs itself** — shrink MW until every site's own SLA holds.

PHYSICS CHECK (REAL AC POWER FLOW)

pandapower `mv_oberrhein`: a real 20 kV German feeder, 179 buses, 2×25 MVA. Base case converges at min V **0.9756 p.u.** with 0 violations. The feeder has **8.2 MW of spare capacity**; our five largest sites need **31.8 MW** — they cannot share one feeder. This is a **siting** problem, not a megawatt problem: 0.5–10 MW edge sites are the right unit.

Inside the loop: with the measured **6.53 MW per feeder** as a hard admission constraint, the same agent gets **2.39×** instead of 4.30× (net $155M vs $280M); all 8 substations pass real AC PF. The delta is the value of real utility ICA data.

REAL hourly 2025 demand and load shape for 12 US balancing authorities — EIA-930, public domain. **12 BA · 3,513 TWh · 8,760 h**

SYNTHETIC site geometry, substation capacity, interconnection timelines (18–36 mo firm, 5–10 mo flexible), battery capex, SLA price multipliers. Every parameter is documented in `sim/assumptions.md`.

Same policy, four operators: CAISO 4.3× · ERCOT 3.41× · PJM 4.07× · MISO 3.88×, zero violations, no per-region tuning.

Run it: `python3 loop/run.py --region CAL --iters 200 --seeds 4 --policy-file rsi_agent/gen_6.py` then `python3 web/build.py results/loop.json`. Live: `python3 web/live_server.py`. Everything offline, no keys, no network.